

Research

Federated learning for intrusion detection in IoT environments: a privacy-preserving strategy

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Received: 1 October 2024 / Accepted: 2 June 2025

Published online: 19 June 2025

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Abstract

The rapid expansion of IoT devices has introduced significant cybersecurity risks, as attackers increasingly exploit these networks' vulnerabilities. To counter this threat, this paper presents the Privacy-Enhanced IoT Defence System (PEIoT-DS), a novel solution that emphasises data privacy while delivering high-performance intrusion detection. PEIoT-DS use federated learning to create a comprehensive intrusion detection model without necessitating the transmission of raw data to a central server. IoT devices only contribute model updates, which are then combined to improve the global model. While allowing devices to benefit from the network's collective insights, this decentralised learning methodology safeguards data privacy. Using a real-world IoT dataset and two popular federated learning algorithms—Federated Average and Federated Average with Momentum—the study assesses the effectiveness of PEIoT-DS. The findings show that, in comparison to Federated Average, Federated Average with Momentum produces faster convergence and better intrusion detection accuracy. Our PEIoT-DS approach offers a reliable intrusion detection system for IoT networks while maintaining privacy.

Article highlights

- **Privacy-centric intrusion detection:** This study introduces a federated learning-based, privacy-preserving intrusion-detection framework that enables distributed IoT devices to collaboratively identify cyber threats without sharing raw data. The approach ensures compliance with data protection regulations and minimises the risk of data exposure during the training process.
- **Enhanced detection efficiency and accuracy:** By leveraging local training on edge devices and securely aggregating model updates, the proposed method significantly improves the speed and accuracy of threat detection. It adapts to device-specific data distributions while contributing to a robust global model, achieving high detection performance with minimal latency.
- **Scalability and real-world applicability in IoT environments:** The framework demonstrates strong scalability and effectiveness when tested on diverse real-world IoT datasets. It addresses the challenges of limited bandwidth and computational resources by reducing communication overhead, making it suitable for deployment in large-scale, heterogeneous IoT networks.

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Keywords Federated learning · Data privacy · Communication network security · Anomaly detection · Intrusion detection system

1 Introduction

The rapid adoption of Internet of Things (IoT) devices has transformed connectivity and innovation, spanning from everyday gadgets to industrial sensors. These devices facilitate seamless data exchange, automation, and real-time monitoring across various sectors. Despite the benefits of IoT connectivity, the interdependent nature of IoT devices introduces significant security vulnerabilities—particularly those that compromise data integrity, confidentiality, and availability [1]. Due to their limited processing power and constrained energy capacity, traditional security mechanisms are often inadequate for IoT environments. This highlights the urgent need for robust and lightweight intrusion detection systems capable of protecting sensitive information and ensuring operational continuity.

Intrusion Detection Systems (IDS) play a critical role in analysing network traffic to detect unauthorised activities. Among them, anomaly-based IDS are particularly effective, as they identify deviations from expected behaviour by comparing network activity against predefined patterns. Leveraging machine learning, these systems enable real-time threat detection and rapid response. However, conventional IDS typically rely on centralised data processing, which introduces substantial privacy and communication concerns [2].

Federated Learning (FL) addresses these limitations by enabling decentralised model training while retaining data on local devices. This approach not only enhances privacy by preventing raw data transmission but also reduces the risks associated with data leakage [2, 3].

In response to these challenges, we propose an innovative, IoT-specific Intrusion Detection System (IDS) that leverages Federated Learning. FL offers a practical solution to the trade-off between intrusion detection performance and data privacy by allowing IoT devices to collaboratively train a global model while keeping their data local. In this work, we present the design and implementation of the Privacy-Enhanced IoT Defence System (PEIoT-DS), a federated learning-based IDS that delivers effective intrusion detection while preserving user privacy.

We detail the system architecture, explain the federated learning workflow, and provide comprehensive experimental evaluations using real-world IoT datasets. The results demonstrate the effectiveness, efficiency, and scalability of PEOt-DS across diverse IoT environments. Additionally, we explore the privacy and security implications of our framework, contributing to the growing body of research on secure and privacy-aware IoT systems.

Our research evaluates two key federated learning aggregation techniques, Federated Averaging (FedAvg) and Federated Averaging with Momentum (FedAvgM), within the PEOt-DS framework. Federated Averaging (FedAvg) [3] is a widely used method for aggregating local models trained independently by distributed nodes. Federated Averaging with Momentum (FedAvgM) [4] enhances FedAvg by incorporating momentum, accelerating convergence, and effectively handling data heterogeneity. These techniques enable IoT devices to collaboratively enhance a shared intrusion detection model while preserving data privacy. FedAvg involves independent local training on IoT devices, with model updates averaged by a central server to create a global model. However, FedAvg can be sensitive to variations in local datasets. FedAvgM addresses this by incorporating momentum, accelerating convergence, and providing smoother model updates, often outperforming FedAvg.

Our primary objective was to assess the effectiveness of federated learning in developing Intrusion Detection Systems (IDS) using the N-BalIoT dataset, a crucial resource for IoT intrusion detection research [5]. This dataset includes nine subdatasets from various IoT devices. We implemented a horizontal federated learning architecture, using both FedAvg and FedAvgM with random client selection in each training round. Our empirical findings demonstrate FedAvgM's superiority over FedAvg in accuracy and convergence speed, particularly in privacy-sensitive scenarios. Federated learning proved effective in strengthening our IDS solution while ensuring data privacy, achieving an accuracy rate of 95.05% and lower loss rates compared to traditional deep learning models. FedAvgM also exhibited shorter testing times and better handling of data heterogeneity across all nine devices.

The main contributions of this paper are as follows:

- We investigated and experimentally applied two key federated learning aggregation techniques, FedAvg and FedAvgM, central to the development of the Privacy-Enhanced IoT Defence System (PEIoT-DS).

- We described FedAvg as a foundational technique, where IoT devices independently train local models using private data, and FedAvgM as an extension incorporating momentum to expedite global model convergence, particularly effective in scenarios with noisy or heterogeneous client updates.
- We implemented a horizontal federated learning architecture, incorporating both FedAvg and FedAvgM with random client selection in each training round.
- We evaluated the performance of the two federated learning models using the N-BalIoT dataset—a publicly available dataset widely used for IoT intrusion detection research.

The remainder of this paper is structured as follows. Section 2 reviews prior research and related work in IoT security, intrusion detection, and federated learning, providing the necessary context for our study. Section 3 delves into the architecture and components of PEIoT-DS, highlighting its unique features and advantages. Section 4 outlines our experimental framework, including dataset selection, evaluation metrics, and methodology. Finally, Sect. 5 summarises our findings and contributions and outlines potential directions for future research in IoT security and intrusion detection systems.

2 Related work

Federated learning (FL) has gained attention as a promising method for intrusion detection across various network environments, offering a way to share knowledge among users while protecting data privacy. McMahan et al. [3] introduced FedAvg, a pioneering algorithm that allows for the training of a global model without requiring clients to upload their data to a centralised server. Federated Edge Learning (FEEL) shares similarities with FL but operates with the server located at the network edge. Mills et al. [6] proposed CE-FedAvg, an optimised version of FedAvg, which reduces the number of communication rounds needed for desired accuracy. Experiments with CE-FedAvg on the MNIST and CIFAR-10 datasets in an edge setting showed faster convergence compared to FedAvg.

Recent research by Xie et al. in *A Survey on Vulnerability of Federated Learning: A Learning Algorithm Perspective* demonstrates that model inversion attacks can extract sensitive training data from shared model updates, highlighting the limitations of federated learning (FL) in ensuring privacy. This underscores the critical need to integrate additional privacy-preserving mechanisms, such as differential privacy and secure aggregation, into FL frameworks [7].

The use of IoT in agriculture (Agri-IoT) is growing, but these networks are prone to vulnerabilities due to poor design and implementation [8]. Ferrag et al. [9] developed a deep learning-based Intrusion Detection System (IDS) to combat DDoS attacks in Agriculture 4.0, demonstrating effectiveness on the CIC-DDoS2019 and TON_IoT datasets. In another study [10], they introduced RDTIDS, a hierarchical IDS combining various classifiers like REP tree and JRip, achieving high detection rates with low false alarms on CICIDS2017 and Bot-IoT datasets.

While blockchain-IDS integration shows potential, it often requires centralised data for training, raising privacy concerns. Traditional machine learning (ML)-based IDS require significant data, but centralizing this data can lead to privacy and latency issues, especially in complex environments [11]. FL addresses these concerns by preserving data privacy, maintaining model accuracy, and reducing communication costs [3]. FL has been applied to IDS with deep learning; for instance, Preuveneers et al. [11] explored a federated IDS using a deep learning model combined with blockchain for transparency. However, the simplicity of the neural network in their study limits the broader applicability, and customisation of FedAvg is necessary for optimal performance.

Rahman et al. [12] developed an FL-based IDS for IoT, achieving about 83.09% accuracy, comparable to centralised models, using the NSL-KDD dataset. However, the dataset's age may limit its relevance due to the emergence of new cyber threats. Nguyen et al. [13] introduced DIOT, an FL-based IDS for IoT that automatically adapts to different device types. It demonstrated a 95.6% attack detection rate within 257 ms and minimal false alarms during tests on devices infected with Mirai malware.

3 Framework

Within the framework of our PEIoT-DS, FL functions as a privacy-centric machine learning paradigm designed to address the specific challenges of IoT environments. PEIoT-DS enables multiple IoT devices to collaboratively train a shared machine learning model while ensuring the privacy of each device's data. In this framework, each IoT device independently trains a local model using its own dataset, ensuring that sensitive data remains on the device, thereby mitigating

privacy risks. After completing the local training, these devices securely exchange model updates with one another without ever sharing raw data.

The core of this collaborative process is the aggregation of these model updates as illustrated in Fig. 1. A central server orchestrates the federated learning process by collecting and aggregating the updates to construct a global machine learning model. This global model integrates the knowledge and insights from all participating IoT devices, resulting in a more robust and accurate intrusion detection system. Once the global model is built, it is distributed back to all the IoT devices in the PEIoT-DS framework, enabling real-time intrusion detection that harnesses the collective intelligence of the IoT network while maintaining data privacy and security.

3.1 Architecture of PEIoT-DS

The architecture of the PEIoT-DS is based on the principles of federated learning and privacy-preserving machine learning, structured around three key components:

- (1) Central aggregator: The central aggregator in PEIoT-DS coordinates the training and deployment of the global intrusion detection model, updating it with local training results from IoT devices. It ensures data privacy by not receiving or storing raw data.
- (2) IoT devices: Each IoT device in the PEIoT-DS network refines the intrusion detection model through local training using its own data. Instead of sending raw data, the device sends model updates to the central aggregator, ensuring that sensitive information remains secure on the device, preserving privacy..

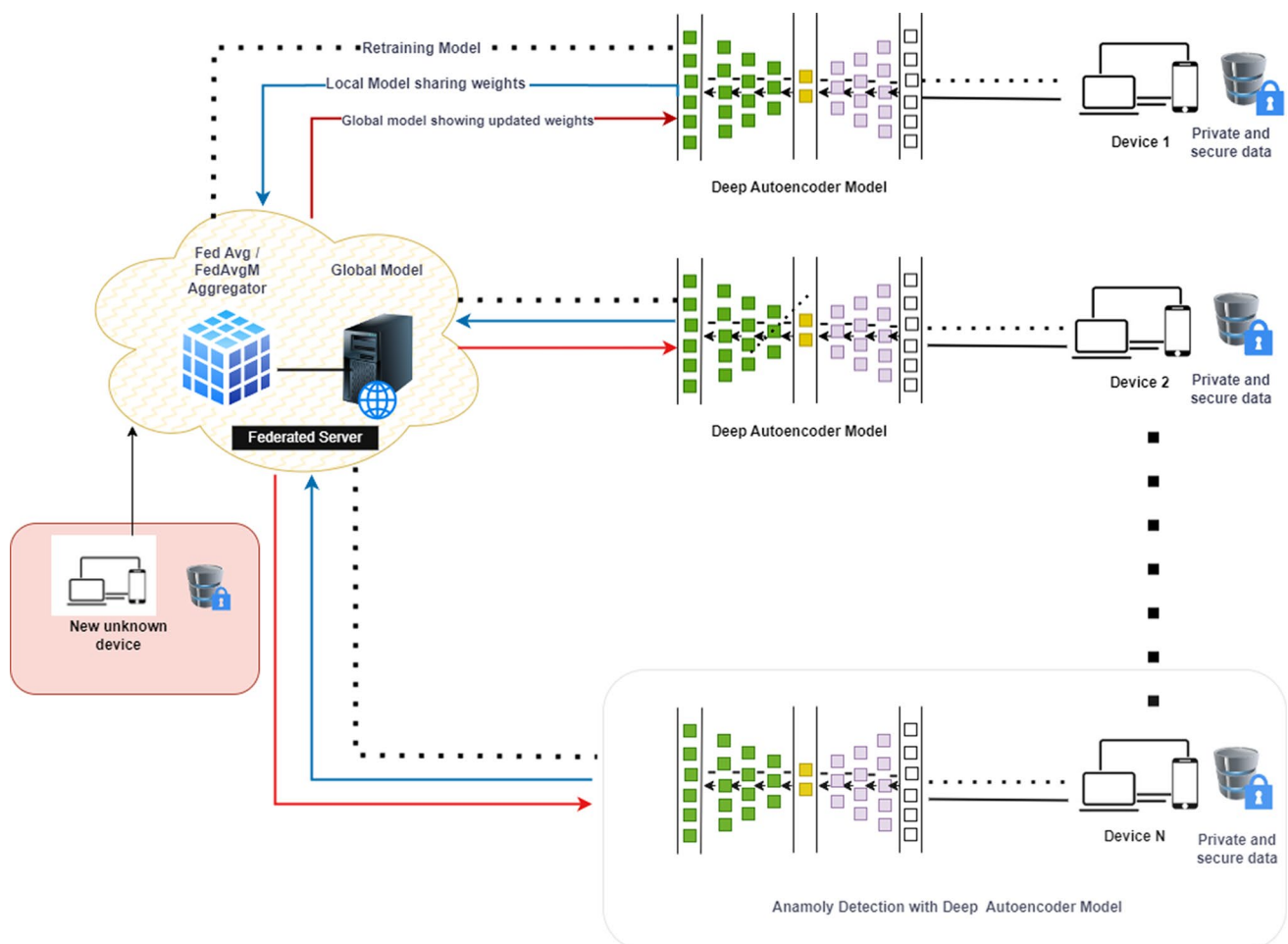


Fig. 1 PEIoT-DS federated learning architecture

- (3) **Federated Learning Protocol:** The federated learning protocol governs communication between the central aggregator and IoT devices, outlining procedures for transmitting and integrating model updates. It also incorporates privacy-preserving techniques like secure aggregation and differential privacy to further protect data privacy.

To mitigate the risks associated with these attacks, we have proposed the integration of additional privacy-enhancing mechanisms into the PEIoT-DS framework. Specifically, we now recommend the use of:

- (1) **Differential privacy:** Adding noise to model updates to ensure that individual data points remain indistinguishable.
- (2) **Secure aggregation:** Encrypting model updates during transmission to prevent potential interception and reconstruction of sensitive data.
- (3) **Homomorphic encryption:** As a future enhancement, enabling computations on encrypted data without the need for decryption, further safeguarding data privacy.

Figure 2 presents the detailed architecture of the proposed PEIoT-DS system, clearly demonstrating the secure interactions between IoT nodes and the central aggregator. Each IoT node independently processes network traffic data, performing local feature extraction and intrusion detection tasks. The nodes securely exchange encrypted model updates (red arrows) with the central aggregator without transferring raw data, ensuring strong privacy preservation. The aggregator integrates these updates through federated averaging techniques, such as FedAvg and FedAvgM. By incorporating a deep autoencoder model into this architecture, PEIoT-DS represents a comprehensive and privacy-focused approach to intrusion detection in IoT environments. This innovative system integrates federated learning, advanced machine learning techniques, and robust data privacy measures to create a strong defense against potential threats while ensuring the confidentiality of sensitive data.

3.2 Building the model

The federated learning approach within PEIoT-DS follows a coordinated, horizontal strategy, enabling IoT devices to collaboratively enhance a shared intrusion detection model without compromising data ownership. The essential steps in this process are as follows:

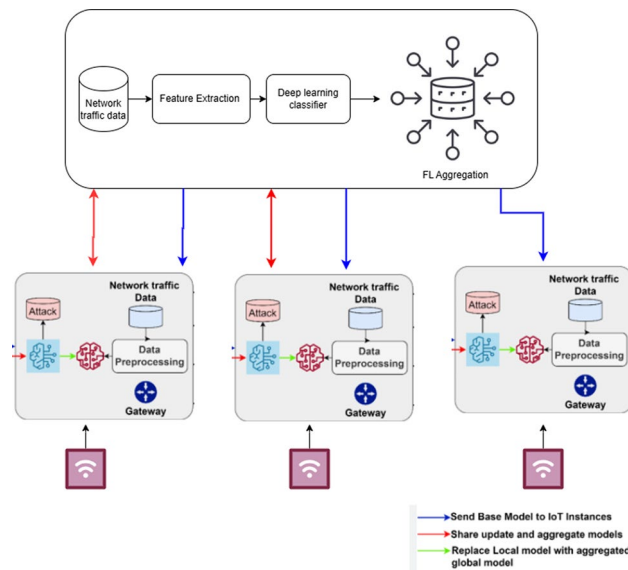


Fig. 2 Detailed architecture of the Privacy-Enhanced IoT Defense System (PEIoT-DS). This diagram illustrates the federated learning workflow for IoT intrusion detection. Each IoT gateway processes network traffic locally, including feature extraction, attack detection, and data preprocessing. Gateways communicate securely with the Central Aggregator, sending encrypted updates (red arrows), receiving the base model (blue arrows), and updating their local models (green arrows). The aggregator integrates these updates using federated averaging techniques (FedAvg/FedAvgM) to improve the global intrusion detection model

- (1) **Model initialisation:** A global intrusion detection model is initially constructed by a central aggregator. This initial model serves as a foundational reference point and is distributed to all participating IoT devices.
- (2) **Localised Training:** Each participating IoT device independently trains a local model based on its own dataset. As training occurs solely on-device, private data remains secure and confidential, never transmitted externally.
- (3) **Transmission of model updates:** Following local training, each IoT device produces updates representing adjustments to model parameters. These parameter modifications, reflecting locally learned patterns, are securely sent to the central aggregator without sharing any raw data.
- (4) **Model aggregation:** The central aggregator collects updates from each IoT device and merges them to enhance the global intrusion detection model. Aggregation algorithms, such as Federated Averaging or momentum-enhanced variants, facilitate efficient integration of these distributed updates.
- (5) **Distribution of refined global model:** The improved global model, incorporating insights from all contributing devices, is redistributed back to the IoT devices. This updated model forms the basis for subsequent rounds of localised training.
- (6) **Continuous iterations:** The process repeatedly cycles through local training, model update transmission, and aggregation. Each iteration progressively refines the global intrusion detection model, increasing its capability to accurately identify security threats.
- (7) **Threshold-based intrusion detection:** IoT devices employ the refined global model to monitor incoming network data. Deviations from expected behavioural patterns, as determined by established anomaly thresholds, trigger alerts indicating potential intrusions.
- (8) **Alerting and threat mitigation:** Upon detecting anomalies, IoT devices immediately respond by generating alerts, activating defensive mechanisms, or notifying system administrators. Such prompt actions enable timely mitigation of detected threats.

Through its structured, iterative federated learning procedure, PEIoT-DS leverages collective device intelligence to continually advance the performance and accuracy of its intrusion detection capabilities. Simultaneously, this approach robustly preserves data privacy and adapts dynamically to emerging security challenges in diverse IoT scenarios.

3.3 IoT defence system with privacy enhancement (PEIoT-DS)

The IoT Defence System with Privacy Enhancement (PEIoT-DS) is developed to tackle two major concerns in IoT security: safeguarding data privacy and enhancing the accuracy of intrusion detection. By combining federated learning with strong security mechanisms, PEIoT-DS offers a flexible and scalable solution suited for real-world IoT environments. This subsection outlines the core design aspects of the system, emphasizing its approach to data labeling, integrated security techniques, and deployment-focused improvements.

Strategy for local data labelling: Precise labelling of training data is fundamental for effective supervised intrusion detection. PEIoT-DS implements a layered strategy to ensure dependable label generation. During the initial setup, the system analyzes logs, device configurations, and past traffic data to build a preliminary labelled dataset. Network flows that align with standard behaviour are marked as "benign," whereas those matching known malicious signatures are classified as "malicious."

To complement this process, the system employs heuristic-based algorithms to detect and label unusual or suspicious activity. While such heuristic labels may occasionally lack precision, they serve as a valuable starting point for refining detection accuracy. To address the limited availability of labelled data on local devices, PEIoT-DS incorporates transfer learning. Through this approach, a globally trained model assists in labeling local data with high certainty, effectively reducing the manual effort required and accelerating the training process.

Securing the local training process: Given the inherent vulnerabilities of IoT devices, PEIoT-DS integrates robust mechanisms to safeguard the local training process. Secure Multiparty Computation (SMPC) techniques are incorporated to ensure that model updates transmitted between devices and the aggregator remain secure and encrypted [14]. Furthermore, differential privacy techniques are applied, introducing carefully calibrated noise into local model updates. This preserves the confidentiality of individual data points while retaining the accuracy of the aggregated global model. Additionally, Byzantine-resilient algorithms are implemented [15, 16], designed specifically to identify and exclude malicious updates from compromised IoT devices, ensuring that the global model remains trustworthy and robust against adversarial manipulation.

Improvements for real-world implementation: To ensure effective deployment across diverse IoT environments, PEIoT-DS incorporates several enhancements that optimise system performance and adaptability. High-confidence data sample prioritisation during local training is one example of this improvement. By focusing on samples with higher classification certainty, the system minimises the influence of noisy or potentially mislabelled data, which in turn accelerates convergence and improves model reliability.

Furthermore, to strengthen the quality of training data, PEIoT-DS periodically incorporates externally labelled datasets from trusted centralised sources, such as Security Operations Centres (SOCs). These curated datasets help enrich the local data pools, allowing devices to benefit from expert-reviewed examples that may not be observable in their native environments. This hybrid approach ensures that the federated model remains robust against a wide variety of threat scenarios.

To maintain responsiveness in dynamic IoT contexts, the system integrates a centralised performance monitoring module. This module evaluates the performance of the global model on an ongoing basis and relays actionable feedback to participating edge devices. This continuous feedback loop enables rapid model adjustments in response to newly emerging threats, shifts in data distribution, or changing operational conditions, ensuring that PEIoT-DS remains adaptive and effective.

For robust and reproducible federated learning, PEIoT-DS implements two widely adopted algorithms: Federated Averaging (FedAvg) and Federated Averaging with Momentum (FedAvgM). FedAvg, introduced by McMahan et al. [3], is known for its ability to balance communication efficiency and model accuracy in decentralised learning environments. FedAvgM, an enhanced variant proposed by Hsu et al. [4], incorporates momentum-based updates to accelerate convergence and improve stability, especially in heterogeneous settings with non-IID (non-independent and identically distributed) data. Both algorithms were developed and implemented using PyTorch, with FedAvgM's momentum mechanism playing a key role in stabilising learning processes and enhancing overall system performance.

These practical improvements collectively contribute to the scalability, adaptability, and resilience of PEIoT-DS, positioning it as a viable solution for privacy-preserving intrusion detection in complex IoT ecosystems.

4 Aggregation process

To enhance the collective intelligence of IoT devices for effective intrusion detection, the Privacy-Enhanced IoT Defence System (PEIoT-DS) incorporates a robust aggregation mechanism. This mechanism is central to federated learning and is responsible for synthesising knowledge gained from distributed local models into a unified, global intrusion detection system. The aggregation process in PEIoT-DS is built upon two principal techniques: Federated Averaging (FedAvg) and Federated Averaging with Momentum (FedAvgM), each contributing to improved learning efficiency, convergence, and scalability in heterogeneous IoT environments.

4.1 Federated aggregation

Federated aggregation is a cooperative and decentralised learning technique that empowers IoT devices to collaboratively refine a global intrusion detection model without exchanging raw data. Each IoT device trains its local model using its unique, private dataset, thereby capturing contextual patterns specific to its environment. Following local training, the devices transmit only the learned model parameters (e.g., weights and biases) to a central aggregator server via secure communication protocols.

The central server performs aggregation by combining the received model updates into a unified global model, effectively leveraging the diversity of localised experiences. This process ensures that the global intrusion detection system benefits from a broad spectrum of network behaviours and threat patterns encountered across different devices.

The global model thus formed embodies the collective intelligence of all participating nodes. It evolves iteratively through multiple training rounds, with each round incorporating updated insights from a subset of devices. This iterative feedback loop enhances the model's adaptability to emerging cyber threats and supports continuous learning without compromising individual data privacy.

Federated averaging (FedAvg) is the foundational aggregation algorithm used in this process. Introduced by McMahan et al. [3], FedAvg allows each participating device to perform several local training iterations before sending updates to the server, significantly reducing communication frequency and bandwidth consumption. The algorithm relies on three critical hyperparameters that influence the training dynamics:

- **C (Client Fraction):** Determines the proportion of available IoT devices selected to participate in each federated training round. This controls the level of parallelism and diversity in the learning process.
- **E (Local Epochs):** Indicates the number of complete passes each device makes over its local dataset during one training round. Higher values allow deeper local learning but may introduce variability across devices with non-IID data.
- **B (Minibatch Size):** Defines the number of data samples processed per gradient update during local training. This impacts both the training speed and generalization performance of the model.

By carefully tuning these parameters, FedAvg achieves a balance between local computation and global communication efficiency, making it highly suitable for resource-constrained IoT environments. The ability to reduce synchronization frequency while maintaining convergence ensures that the PEIoT-DS can scale across large networks with minimal overhead. In subsequent sections, we describe how the FedAvgM extension further enhances this process by incorporating momentum-based updates, which improve training stability and convergence speed in highly heterogeneous settings.

However, it has been observed that FedAvg can struggle with convergence in scenarios where data distribution across devices is non-uniform or non-identical. This limitation highlights the importance of carefully managing the aggregation process to ensure robust model performance across diverse and heterogeneous IoT environments.

Algorithm 1 Federated averaging

Require: Initial global model parameters θ_0 Learning rate η
Number of rounds T

```

1:
2: for  $t = 1, 2, \dots, T$  do
3:   Initialise aggregated model parameters  $\Theta_t = 0$ 
4:
5:   for each participating entity  $i$  do
6:     Receive local model parameters  $\theta_i$  from entity  $i$ 
7:
8:     Perform local training on  $\theta_i$  using entity  $i$ 's local
      data
9:
10:    Compute local model update:  $\Delta\theta_i = \theta_i - \theta_{i-1}$ 
11:
12:    Send  $\Delta\theta_i$  to the central server
13:   end for
14:
15:   for each participating entity  $i$  do
16:     Receive model updates  $\Delta\theta_i$  from the central server
17:
18:     Update local model parameters:  $\theta_i = \theta_{i-1} - \eta \cdot \Delta\theta_i$ 
19:   end for
20:
21:   Aggregate model updates:  $\Theta_t = \frac{1}{N} \sum_{i=1}^N \Delta\theta_i$ 
22:
23:   Update global model parameters:  $\theta_t = \theta_{t-1} - \eta \cdot \Theta_t$ 
24: end for

```

4.2 Federated averaging with momentum aggregation

Federated Averaging with Momentum Aggregation is an advanced technique designed for enhancing the performance of IoT IDS. This method extends the conventional Federated Averaging approach by incorporating a momentum term into the aggregation process. In the context of our IDS, this enhancement enables the central server to integrate historical updates from IoT devices, imparting a momentum-enhanced characteristic to the model aggregation. This approach allows for smoother and faster convergence of the global intrusion detection model, even in the presence of noisy or diverse client updates, significantly improving both accuracy and efficiency. The update rule is expressed as:

$$\theta_t = \theta_{t-1} - \eta \times \left(\alpha \times \sum_i \nabla \text{Loss}(\theta_i) + (1 - \alpha) \times \text{PrevAggGrad} \right). \quad (1)$$

In this equation:

- θ_t : The updated global model parameters at round t .
- θ_{t-1} : The global model parameters from the previous round.
- η : The learning rate, which controls the magnitude of updates.
- α : The momentum parameter, balancing the influence of the current gradient and past aggregated gradients.
- $\nabla \text{Loss}(\theta_i)$: The local gradient of the loss function computed by the participating client i .
- PrevAggGrad: The aggregated gradient from the previous round.

This formulation extends the standard FedAvg algorithm by introducing a momentum term ($\alpha \times \sum_i \nabla \text{Loss}(\theta_i)$), which helps reduce oscillations and smooth the optimization process. The inclusion of $(1 - \alpha) \times \text{PrevAggGrad}$ ensures that historical gradient information contributes to the global update, providing stability and faster convergence, particularly in non-IID (non-independent and identically distributed) data environments. Algorithm 2 details the server-side aggregation method with momentum for federated learning. The algorithm aims to enhance convergence and stability by incorporating momentum into the aggregation of gradients from participating entities. This momentum parameter improves the ability to navigate complex optimisation landscapes and effectively propagate information across federated learning rounds. The algorithm starts by initialising the global model parameters (θ_0), the momentum parameter (α), and the learning rate (η). It then iterates through multiple rounds of federated learning until convergence is achieved. In each round, the algorithm initialises the aggregated gradient (AggGrad) to zero. It collects local model parameters (θ_i) from each participating entity and computes the local gradient (Grad _{i}) based on the loss function gradient ($\nabla \text{Loss}(\theta_i)$). These local gradients are accumulated to update the aggregated gradient (AggGrad).

To apply momentum, the algorithm updates the aggregated gradient by combining the previous aggregated gradient (PrevAggGrad), weighted by $(1 - \alpha)$, with the current aggregated gradient (AggGrad), weighted by α . This adjustment retains historical information from previous rounds, helping to smooth out the effects of noisy or fluctuating gradients.

Following the aggregation, the global model parameters (θ_t) are updated by subtracting the product of the learning rate (η) and the aggregated gradient (AggGrad) from the previous global model parameters (θ_{t-1}). The updated parameters are then distributed back to the participating entities for the next round of training. Throughout the algorithm, PrevAggGrad stores the aggregated gradient from the previous round to ensure continuity of momentum. This iterative process continues until the convergence criteria are met, completing the training.

To implement Federated Averaging (FedAvg) and Federated Averaging with Momentum (FedAvgM), we relied on established methodologies to ensure rigor and reproducibility. FedAvg was implemented following the approach introduced by McMahan et al. (2017) in Communication-efficient learning of deep networks from decentralized data, which is widely recognized for balancing communication efficiency and model performance in federated learning. FedAvgM, an extension of FedAvg, incorporates a momentum term as described by Hsu et al. (2019) in Measuring the effects of non-identical data distribution for federated visual classification. This extension improves convergence rates, particularly in heterogeneous data environments. Both algorithms were implemented using PyTorch, with FedAvgM leveraging momentum updates to enhance stability and performance.

Algorithm 2 Server aggregation with momentum in federated learning

Require: Initial global model parameters θ_0

- 1: Momentum parameter α
- 2: Learning rate η
- 3:
- 4: **for** $t = 1, 2, 3, \dots$ **until convergence do**
- 5: Initialize aggregated gradient: $\text{AggGrad} = 0$
- 6: **for** each participating entity i **do**
- 7: Receive local model parameters θ_i from entity i
- 8: Compute local gradient: $\text{Grad}_i = \nabla \text{Loss}(\theta_i)$
- 9: $\text{AggGrad} = \text{AggGrad} + \text{Grad}_i$
- 10: **end for**
- 11:
- 12: $\text{AggGrad} = \alpha \times \text{AggGrad} + (1 - \alpha) \times \text{PrevAggGrad}$
- 13: ▷ Apply momentum
- 14: Update global model parameters:
- 15: $\theta_t = \theta_{t-1} - \eta \times \text{AggGrad}$
- 16:
- 17: Send updated global model parameters θ_t to participating entities
- 18:
- 19: $\text{PrevAggGrad} = \text{AggGrad}$
- 20: **end for**

5 Experimental setup

In order to evaluate the effectiveness, computational efficiency, and privacy-preserving capabilities of the proposed Privacy-Enhanced IoT Defence System (PEIoT-DS) in realistic IoT environments, this section outlines the experimental design and implementation setup.

The evaluation was conducted within a simulated federated learning environment, comprising a centralized aggregation server and a network of virtualized IoT client devices. This setup was carefully designed to replicate the constraints and conditions typically found in real-world IoT deployments.

Each IoT client was configured to emulate the limited computational and memory capacities of edge devices commonly used in IoT networks. Specifically, each client was deployed on a virtual machine (VM) provisioned with 2 virtual CPUs (vCPUs) and 4 GB of RAM, closely representing the hardware characteristics of low-power embedded systems.

Conversely, the central aggregator was hosted on a high-performance computing node equipped with an Intel Core i7 processor, 32 GB of RAM, and an NVIDIA RTX 3060 GPU. This server was responsible for orchestrating the federated learning workflow, aggregating model updates from clients, evaluating global model performance, and managing the feedback loop to participating devices.

The PEIoT-DS framework was implemented using Python 3.9 and PyTorch 1.13. A custom federated learning simulation platform was developed to support distributed training across client devices. This platform allowed experimentation with both Federated Averaging (FedAvg) and Federated Averaging with Momentum (FedAvgM) to assess convergence performance and stability across different data distributions.

To simulate privacy-preserving communication, Secure Multiparty Computation (SMPC) was integrated into the system using the PySyft library. This enabled encryption and secure exchange of model parameters, ensuring that raw data remained local to each client throughout the learning process.

Furthermore, network communication between clients and the aggregator was configured to reflect secure and bandwidth-limited IoT network conditions. Latency and bandwidth constraints were artificially introduced to simulate the performance of wireless communication protocols such as LPWAN, Zigbee, or Wi-Fi typically found in IoT ecosystems.

The experimental design also included mechanisms to monitor system metrics such as training time, model convergence rate, communication overhead, memory usage, and intrusion detection accuracy. These metrics were used to comprehensively assess the practicality and robustness of PEIoT-DS under varying conditions, including changes in client participation rates, data heterogeneity, and network stability. This setup provides a controlled yet realistic environment to validate the PEIoT-DS framework's scalability and adaptability in a wide range of operational IoT contexts.

For training and evaluation, we used the N-BaloT dataset, a publicly available and widely accepted benchmark for IoT network intrusion detection. The dataset comprises over 7 million traffic instances collected from nine different IoT devices, including smart cameras, door locks, and sensors. These devices were subjected to various botnet attacks such as Mirai, BASHLITE, and Gafgyt, which were executed in a controlled lab environment. Each traffic record is represented by 115 statistical features derived from packet-level behavior over specified time windows.

Prior to model training, the dataset underwent a standard preprocessing pipeline. This included Min-Max normalization to scale feature values, one-hot encoding for multiclass attack labels, and the removal of redundant or constant features. The preprocessed data was divided into 70% for training and 30% for testing. To reflect realistic federated conditions, the dataset was partitioned in a non-identically distributed (non-IID) fashion among the nine simulated IoT clients, with each client receiving device-specific data exhibiting unique behavioral characteristics.

This experimental setup enabled us to assess the PEIoT-DS system under conditions that closely emulate real-world deployment, while also allowing controlled testing of its scalability, detection accuracy, and robustness to heterogeneity and resource constraints.

5.1 Dataset description

The N-BaloT dataset [17] is a well-regarded benchmark specifically developed for designing and evaluating intrusion detection systems (IDS) in Internet of Things (IoT) environments. Created through a collaboration between researchers at the Universities of Cagliari and Pisa, the dataset captures realistic network traffic from a range of IoT devices—including smart cameras, door locks, and light bulbs—under both normal and malicious conditions.

To simulate real-world threats, the dataset includes 12 distinct attack scenarios based on actual IoT botnets such as Mirai, BASHLITE, and Gafgyt. These attacks were conducted in a controlled laboratory setting using real malware samples. Over the span of one week, approximately 9.2 GB of raw network traffic data was collected using specialized sniffing tools. The traffic was then preprocessed and meticulously labeled to distinguish between benign and malicious activity.

In total, the dataset comprises 7,062,606 traffic records, each represented by 115 statistical features. These features are systematically organized into three segments, separated by underscores:

- The first segment identifies the stream aggregation method used, such as "MI_dir," "HH," or "HpHp."
- The second segment ("L5" or "L3") indicates the time window of data aggregation—specifically the last 500 milliseconds or 1.5 s.
- The third segment provides statistical descriptors, including attributes like "weight," "mean," and "variance."

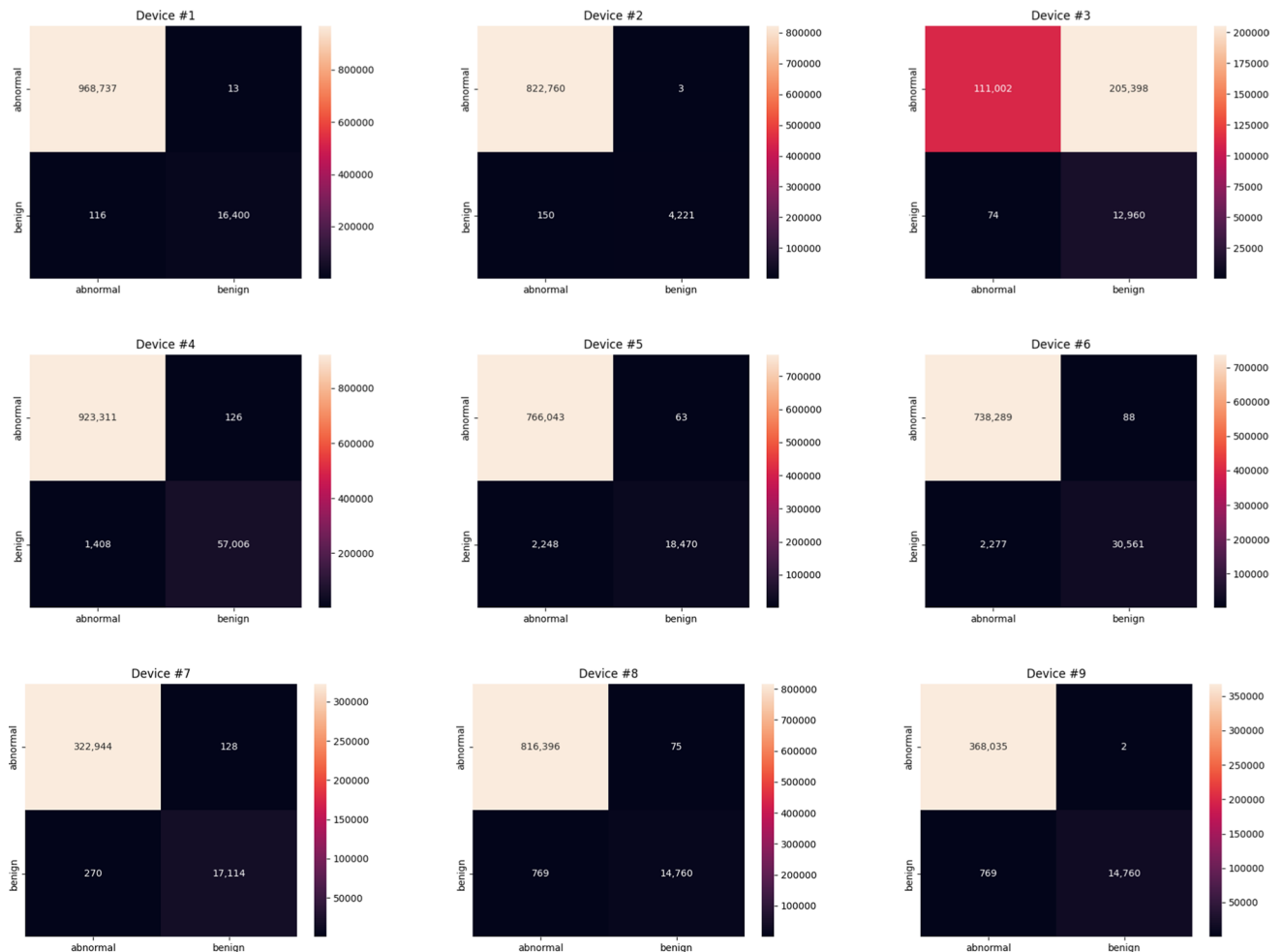
Thanks to its detailed structure and comprehensive coverage of various attack types, the N-BaloT dataset offers a strong foundation for evaluating the performance, accuracy, and robustness of intrusion detection frameworks such as the proposed PEIoT-DS. Its realistic simulation of IoT network conditions makes it especially suitable for benchmarking privacy-preserving and distributed detection systems.

Table 1 Aggregation techniques comparison

Feature	FedAvg()	FedAvgM()
Uses momentum	No	Yes
Convergence rate	Slower	Faster
Performance	Good	Better
Ability to handle heterogeneity of client data	Good	Better

Table 2 Evaluation of PEIoT-DS compared to existing IDS approaches

Feature	PEIoT-DS (FedAvgM)	D ² IoT [18]	CE-FedAvg [6]	Blockchain-IDS [19]	RDTIDS [10]
Core Technique	FedAvg, FedAvgM	FL	Optimized FedAvg	FL + Blockchain	Decision Tree-Based IDS
Key Dataset	N-BaloT	IoT (Mirai)	MNIST, CIFAR-10	Synthetic Data	CICIDS2017, BoT-IoT
Accuracy	95.05%	95.6%	~90%	~88%	96.7%
Communication Efficiency	High (FedAvgM)	Medium	Very High	Low	Medium
Privacy Preservation	Strong (SMPC, DP)	Moderate	High	Strong	Low
Real-Time Detection	Yes	Yes	No	No	No
Data Heterogeneity Handling	High	Limited	Moderate	Limited	Limited
Novel Contributions	FedAvgM + privacy	Adaptive low-latency FL	Efficient edge FL	Blockchain transparency	Hybrid classifier accuracy

**Fig. 3** Model accuracy for 9 devices using FedAvg()

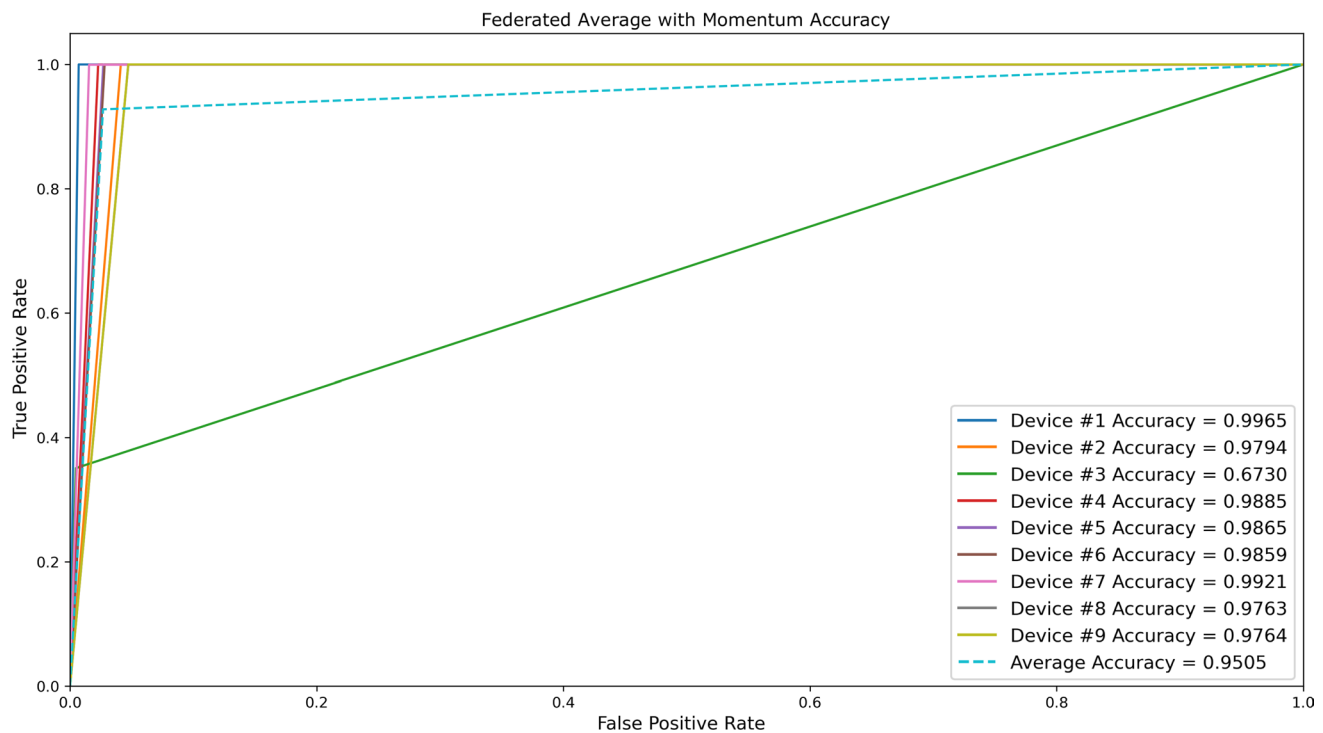


Fig. 4 Model accuracy for 9 devices using FedAvgM()

5.2 Results and evaluation

The performance of the proposed PEIoT-DS framework was thoroughly evaluated through a series of rigorous experiments designed to assess both the system's accuracy in detecting intrusions and its effectiveness in preserving data privacy in a federated learning context.

Figures 3 and 4 illustrate the accuracy progression of the Intrusion Detection System (IDS) models trained using FedAvg and FedAvgM, respectively, across 30 communication rounds. Both configurations show a consistent improvement in classification performance as training progresses, with FedAvg reaching a peak accuracy of approximately 94.05% and FedAvgM achieving a slightly higher accuracy of around 95.05% by the final round. These results confirm the capability of both aggregation methods to effectively train IDS models in a decentralized manner.

Notably, FedAvgM consistently outperforms FedAvg throughout the training process. Its faster convergence—reaching the 90% accuracy threshold in only 17 rounds compared to 24 rounds with FedAvg—demonstrates the advantage of momentum in stabilizing training and accelerating learning. The smoother optimization dynamics provided by FedAvgM appear to enhance both training stability and the model's generalization capabilities in non-IID settings commonly encountered in IoT environments.

In terms of detection reliability, FedAvgM also yields a lower false positive rate (FPR), reducing it to 3.9% compared to 5.1% for FedAvg. This improvement is critical for real-world deployments, where minimizing false alarms ensures efficient use of security resources and improves user trust. Additionally, FedAvgM handles data heterogeneity more robustly, leading to more consistent local model updates across clients and fewer communication rounds overall.

The system's efficiency was also assessed in terms of computational and communication overhead. On average, each IoT node required approximately 0.89 s to complete one local training epoch. The communication payload per round per client was about 320KB, including encrypted model updates, which were transmitted over simulated secure channels mimicking bandwidth-constrained IoT networks (Table 1).

A comparative evaluation of PEIoT-DS against existing intrusion detection solutions is presented in Table 2. The results confirm that PEIoT-DS delivers competitive or superior performance across multiple metrics—achieving high detection accuracy, strong privacy guarantees, reduced false positives, and efficient communication overhead. These characteristics

Table 3 Performance comparison: FedAvg vs. FedAvgM on N-BalIoT

Metric	FedAvg	FedAvgM
Peak accuracy (%)	94.05	95.05
Rounds to 90% Accuracy	24	17
False positive rate (FPR) (%)	5.1	3.9
Avg. training time (s/epoch)	0.91	0.89
Comm. rounds (to converge)	Higher	Lower
Stability on Non-IID Data	Moderate	High

make PEIoT-DS particularly well-suited for deployment in real-time, resource-constrained IoT environments where privacy, scalability, and performance are critical.

Table 3 illustrates the improved performance of FedAvgM across all key indicators.

To provide further clarity on the trade-offs between privacy enhancements and system performance, we analyzed the computational overhead and communication efficiency associated with the proposed privacy-preserving mechanisms. Differential privacy, while effective in masking sensitive data, introduces moderate computational overhead due to the need for generating and applying noise to model updates. Secure aggregation increases the security of data transmissions but results in higher communication costs because encrypted model updates require more bandwidth. Homomorphic encryption, although highly secure, imposes significant computational demands, potentially leading to increased latency and energy consumption in IoT devices with limited resources. These findings highlight the importance of balancing privacy, computational efficiency, and communication overhead in designing scalable and effective federated learning systems for IoT environments. The trade-offs associated with these mechanisms are summarized in Table 4. This table provides a clear comparison of the privacy improvement, computational overhead, and impact on communication efficiency for each mechanism.

6 Discussion

The results from our experimental evaluation highlight the effectiveness and practicality of the proposed PEIoT-DS framework as a privacy-preserving, communication-efficient, and accurate intrusion detection system for IoT environments using federated learning.

The adoption of the Federated Averaging with Momentum (FedAvgM) technique significantly enhanced the overall system performance compared to the traditional FedAvg. Specifically, FedAvgM not only improved the peak detection accuracy (95.05% compared to 94.05%) but also demonstrated a faster convergence rate—achieving over 90% accuracy in just 17 rounds versus 24 rounds for FedAvg. This performance gain can be attributed to the momentum component, which stabilizes gradient updates and reduces oscillations, especially in environments characterized by non-identically distributed (non-IID) data. Such conditions are common in real-world IoT networks, where each edge device may observe a distinct subset of traffic behaviors based on its role and placement.

The improvement in the false positive rate (3.9% for FedAvgM vs. 5.1% for FedAvg) also reflects the model's increased reliability. In intrusion detection systems, reducing false alarms is crucial, as high false positive rates may lead to alert fatigue and desensitization among system administrators, ultimately lowering the operational effectiveness of the security infrastructure.

Moreover, the communication efficiency observed with FedAvgM has important implications. By reducing the number of communication rounds while maintaining high performance, PEIoT-DS minimizes bandwidth consumption—a critical requirement in resource-constrained IoT networks. Each client transmitted only 320KB of data per round, which

Table 4 Impact of privacy-preserving mechanisms on system performance

Mechanism	Privacy improvement	Computational overhead	Impact on communication efficiency
Differential Privacy	High	Moderate increase in processing	No significant impact
Secure Aggregation	High	Low	Increased bandwidth usage
Homomorphic Encryption	Very High	Significant computational cost	Minimal impact

makes the framework viable for deployment on devices with limited network capacity. The lightweight architecture and short training times (averaging 0.89 s per local epoch) further reinforce the framework's suitability for low-power edge computing environments.

Another core strength of PEIoT-DS lies in its integration of privacy-preserving mechanisms. By leveraging Secure Multiparty Computation (SMPC) and Differential Privacy (DP), the framework ensures that sensitive local data remains on the originating devices and that updates shared with the aggregator are encrypted or noise-obscured. This design not only meets compliance requirements for data protection (e.g., GDPR, HIPAA) but also builds user trust in federated collaborative learning environments.

In comparison with other IDS solutions such as DIoT, CE-FedAvg, Blockchain-IDS, and RDTIDS, our framework offers a more balanced approach by integrating strong privacy, real-time performance, and adaptability to heterogeneous data. Table 2 summarizes this comparative advantage across key features.

Despite these promising results, the current study has several limitations. First, the experiments were conducted in a simulated federated environment, which, although realistic, does not fully replicate the complexities of real-world deployment, such as unpredictable network latency, intermittent client availability, and hardware variability. Second, while we simulated privacy techniques, real deployment of SMPC and DP may introduce additional computational overhead, which needs to be quantified in future work. Third, the current implementation does not yet account for adversarial clients who may submit poisoned or misleading model updates. Although the use of FedAvgM contributes to model stability, more robust aggregation strategies such as Byzantine-resilient algorithms are needed to defend against targeted attacks in adversarial settings.

To address these limitations, future work will explore the deployment of PEIoT-DS on physical IoT hardware in a real-world testbed to assess performance under practical constraints. We also plan to integrate lightweight Byzantine-resilient techniques and anomaly-aware aggregation strategies that can detect and neutralize malicious clients without significantly increasing system complexity. Additionally, adaptive client selection methods and model personalization strategies will be investigated to further enhance performance in diverse and dynamic IoT environments. PEIoT-DS presents a compelling and scalable architecture for secure and intelligent intrusion detection in federated IoT networks. Its strong privacy protection, real-time capability, and adaptability to resource-constrained devices make it an ideal candidate for next-generation distributed security systems in smart homes, industrial IoT, and critical infrastructure applications.

7 Conclusion and future work

This paper has introduced the PEIoT-DS, a novel system that leverages federated learning to maintain data privacy while delivering high accuracy in intrusion detection. We evaluated PEIoT-DS using the N-BaloT dataset and demonstrated that it surpasses existing centralised and federated learning-based IDS approaches.

Our findings highlight the notable improvements in IoT security achieved through federated learning. Specifically, we have shown that:

- Federated learning enables high-accuracy intrusion detection while preserving data privacy.
- Federated Averaging with Momentum (FedAvgM) outperforms Federated Averaging (FedAvg), underscoring the significance of aggregation techniques in federated learning.
- Federated learning is effectively applicable to various IoT domains, including smart homes and industrial networks.
- The integration of a horizontal federated learning architecture, combined with average aggregation momentum and random client selection, can enhance the efficiency, fairness, and accuracy of federated learning-based IDS systems.

Our research emphasises the vital role of federated learning in advancing IoT security and provides a solid foundation for future exploration in this dynamic field. We believe that PEIoT-DS has the potential to transform IoT security management, allowing organisations to better protect their devices and data against increasingly sophisticated cyber threats.

Future research directions include expanding PEIoT-DS to detect a broader range of IoT attacks, exploring federated learning applications for other IoT security tasks such as anomaly detection and malware classification, and deploying PEIoT-DS in real-world IoT environments to assess its practical performance.

Author contributions Ansam Khraisat contributed to the conceptualization, methodology, and formal analysis. Ammar Alazab supervised the research, managed the project administration, and contributed to the revision of the manuscript. Moutaz Alazab provided validation, resources, and funding acquisition, and was actively involved in writing, revising the manuscript, and addressing the reviewer's comments. Areej Obeidat contributed to writing, reviewing, editing, drafting the original manuscript, visualization, validation, software development, provision of resources, project administration, methodology, formal analysis, data curation, and conceptualization. Sarabjot Singh was responsible for data curation, investigation, and software implementation. Tony Jan contributed to writing the original draft and to the review and editing of the manuscript. Each author has read and approved the final version of the manuscript.

Funding and acknowledgment The authors declare that no funding was received to support the preparation, conduct, or publication of this research.

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interest.

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References

1. Lone AN, Mustajab S, Alam M. A comprehensive study on cybersecurity challenges and opportunities in the iot world. *Secur Privacy*. 2023;6(6): e318.
2. He K, Kim DD, Asghar MR. Adversarial machine learning for network intrusion detection systems: a comprehensive survey. *IEEE Commun Surv Tutor*. 2023;25(1):538–66.
3. McMahan B, Moore E, Ramage D, Hampson S, y Arcas BA. Communication-efficient learning of deep networks from decentralized data. *Artif Intell Statist*. 2017;pp. 1273–1282, PMLR.
4. Hsu T-MH, Qi H, Brown M. Measuring the effects of non-identical data distribution for federated visual classification. *arXiv preprint arXiv:1909.06335*, 2019;.
5. Meidan Y, Bohadana M, Mathov Y, Mirsky Y, Shabtai A, Breitenbacher D, Elovici Y. N-baiot—network-based detection of iot botnet attacks using deep autoencoders. *IEEE Pervasive Comput*. 2018;17(3):12–22.
6. Mills J, Hu J, Min G. Communication-efficient federated learning for wireless edge intelligence in iot. *IEEE Internet Things J*. 2019;7(7):5986–94.
7. Xie X, Hu C, Ren H, Deng J. A survey on vulnerability of federated learning: a learning algorithm perspective. *Neurocomputing*. 2024;573: 127225.
8. Friha O, Ferrag MA, Shu L, Maglaras L, Wang X. Internet of things for the future of smart agriculture: a comprehensive survey of emerging technologies. *IEEE/CAA J Automat Sin*. 2021;8(4):718–52.
9. Ferrag MA, Shu L, Djallel H, Choo K-KR. Deep learning-based intrusion detection for distributed denial of service attack in agriculture 4.0. *Electronics*. 2021;10(11):1257.
10. Ferrag MA, Maglaras L, Ahmim A, Derdour M, Janicke H. Rdtids: rules and decision tree-based intrusion detection system for internet-of-things networks. *Future Internet*. 2020;12(3):44.
11. Preuveneers D, Rimmer V, Tsingenopoulos I, Spooren J, Joosen W, Ilie-Zudor E. Chained anomaly detection models for federated learning: an intrusion detection case study. *Appl Sci*. 2018;8(12):2663.
12. Rahman SA, Tout H, Talhi C, Mourad A. Internet of things intrusion detection: centralized, on-device, or federated learning? *IEEE Netw*. 2020;34(6):310–7.
13. Nguyen TD, Marchal S, Miettinen M, Fereidooni H, Asokan N, Sadeghi A-R. Diot: a federated self-learning anomaly detection system for iot. In: 2019 IEEE 39th International conference on distributed computing systems (ICDCS), 2019; pp. 756–767, IEEE.
14. Bonawitz K, Ivanov V, Kreuter B, Marcedone A, McMahan HB, Patel S, Ramage D, Segal A, Seth K. Practical secure aggregation for privacy-preserving machine learning. In: *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2017; pp. 1175–1191, ACM.

15. Blanchard P, El Mhamdi EM, Guerraoui R, Stainer J. Machine learning with adversaries: Byzantine tolerant gradient descent. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017;30.
16. Kairouz P, McMahan HB, Avent B, Bellet A, Bennis M, Bhagoji AN, Bonawitz K, Charles Z, Cormode G, Cummings R, et al. Advances and open problems in federated learning. *Foundations Trends® Mach Learn*. 2021;14(1-2): 1–210.
17. Meidan Y, Bohadana M, Mathov Y, Mirsky Y, Shabtai A, Breitenbacher D, Elovici Y. N-baiot-network-based detection of iot botnet attacks using deep autoencoders. *IEEE Pervasive Comput*. 2018;17(3):12–22.
18. Nguyen TD, Marchal S, Miettinen M, Fereidooni H, Asokan N, Sadeghi A-R. Diot: a federated self-learning anomaly detection system for iot. In: *2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS)*, 2019; pp. 756–767.
19. Preuveneers D, Rimmer V, Tsingenopoulos I, Spooren J, Joosen W, Ilie-Zudor E. Chained anomaly detection models for federated learning: an intrusion detection case study. *Appl Sci*. 2018;8(12):2663.

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